

**WT : EXPERIMENT 1****Bluetooth :**

Bluetooth is a short-range wireless technology primarily used for personal area networks (PAN). It typically has a communication range of up to 100 meters, although this can vary depending on the Bluetooth version and environment. The cost of Bluetooth modules is low, and it is widely used in smartphones, headphones, and other consumer electronics. Propagation delay is low, often in the millisecond range, which makes it suitable for real-time communication. Bluetooth's power consumption is generally low, especially in Bluetooth Low Energy (BLE), which is optimized for battery-powered devices. Its throughput varies from 1 Mbps in older versions to about 2-3 Mbps in the latest Bluetooth 5.x standards.

**Wi-Fi :**

Wi-Fi is a wireless networking technology used to provide high-speed internet access over a range of distances, typically 100 to 150 meters. The cost of Wi-Fi modules can range from moderate to high depending on the generation and technology, with newer versions (like Wi-Fi 6) being more expensive. Wi-Fi has a moderate propagation delay (milliseconds) and can be affected by factors such as distance and obstacles. It consumes moderate power, but newer standards (like Wi-Fi 6) are designed to be more power-efficient. Throughput is one of Wi-Fi's key strengths, with speeds ranging from 54 Mbps in Wi-Fi 4 to up to 9.6 Gbps in Wi-Fi 6.

**Zigbee (XB) :**

Zigbee is a low-power, low-data-rate wireless communication technology mainly used for Internet of Things (IoT) applications. It has a communication range of around 100 meters, which can be extended using mesh networking. The cost of Zigbee devices is low, making it an attractive option for low-cost IoT implementations. Its propagation delay is relatively low, but slightly higher compared to Bluetooth. Zigbee's power consumption is very low, which is ideal for battery-operated devices in applications like home automation. The throughput is also low, typically ranging from 20 kbps to 250 kbps, suitable for simple data transmissions.

### **LoRa :**

LoRa (Long Range) is a wireless technology designed for long-range communication, particularly in IoT and low-power applications. It offers a communication range of up to 15 kilometers in rural environments, making it ideal for wide-area sensor networks. The cost of LoRa technology is low, which makes it an attractive option for large-scale deployments. Propagation delay is high due to its focus on long-range, low-data-rate communication. Its power consumption is very low, making it ideal for devices that need to run on batteries for extended periods. LoRa has a low throughput range, typically from 0.3 kbps to 27 kbps, as it is optimized for small, infrequent data packets.

### **5G :**

5G is the latest generation of cellular technology, offering extremely high speeds, low latency, and massive device connectivity. The communication range is typically 1-2 kilometers in urban environments, but it can vary based on infrastructure. 5G is known for its high cost due to the advanced infrastructure needed for deployment. The propagation delay in 5G is very low, typically between 1-10 milliseconds, enabling real-time applications like autonomous vehicles and high-quality video streaming. Power consumption is moderate to high, although 5G is optimized for energy efficiency in certain use cases. Throughput is one of 5G's standout features, with speeds ranging from 100 Mbps to 10 Gbps, offering extremely fast data transfer.

### **NFC (Near Field Communication) :**

NFC is a very short-range communication technology used primarily for contactless payments and data exchange between devices placed within a few centimeters of each other. The range of NFC is very short, typically up to 10 centimeters. Its cost is low, and it is integrated into many modern smartphones and payment systems. The propagation delay in NFC is extremely low, often being near-instantaneous. NFC devices consume very little power, as they only communicate over short distances and for brief moments. The throughput is low, with speeds generally reaching up to 424 kbps, which is suitable for applications like mobile payments and ticketing.

### **Infrared (IR) :**

Infrared (IR) is a short-range wireless communication technology commonly used in remote controls, data transfer between devices, and other consumer electronics. The communication range is typically limited to 5 meters, though this can vary depending on the strength and alignment of the infrared signals. Infrared modules are low in cost and are often used in budget electronics. Propagation delay is moderate, typically in the milliseconds range, and can be affected by environmental factors such as obstructions. IR has low power consumption, making it ideal for small, battery-operated devices. Its throughput is also low, generally up to 4 Mbps, which is sufficient for simple commands and control signals.

### **Satellite Communication :**

Satellite communication enables long-distance communication by transmitting signals to and from satellites orbiting the Earth. It provides global coverage, making it ideal for remote areas where terrestrial communication infrastructure is not available. However, the cost is high due to the need for satellite infrastructure and maintenance. The propagation delay in satellite communication is significant, often ranging from 500 milliseconds to over a second due to the long distances the signals must travel. Power consumption is also high, as both satellites and ground stations require substantial energy. Throughput varies depending on the satellite system, but it typically ranges from 1 Mbps to 100 Mbps, depending on the network type and satellite technology.

### **Cellular Communication (GSM, CDMA, 3G, 4G) :**

Cellular communication refers to various generations of mobile network technologies, from GSM (2G) to the latest 4G LTE. GSM typically has a range of several kilometers, while newer generations like 4G can cover distances up to 50 kilometers in rural areas. The cost of cellular communication can be moderate to high, depending on the technology used. Propagation delay varies by generation, with 4G having a delay of around 30-100 milliseconds, while older technologies like GSM have higher delays. Power consumption varies, with 4G being more energy-efficient than previous generations. Throughput differs across generations: GSM offers 56-114 kbps, CDMA ranges from 144 kbps to 2 Mbps, 3G provides 384 kbps to 42 Mbps, and 4G can reach speeds of 100 Mbps to 1 Gbps.

**Comparison Table**

| Parameter              | Bluetooth         | Wi-Fi                    | Zigbee (XB)         | LoRa                | 5G                       | NFC              | Infrared         | Satellite                  | Cellular (GSM, CDMA, 3G, 4G)    |
|------------------------|-------------------|--------------------------|---------------------|---------------------|--------------------------|------------------|------------------|----------------------------|---------------------------------|
| Range of Communication | 100 meters        | 100-150 meters           | 100 meters          | Up to 15 km         | 1-2 km (urban)           | 10 cm            | 5 meters         | Global coverage            | 1-50 km (depends on tech)       |
| Cost                   | Low (₹100 - ₹500) | Moderate (₹500 - ₹2,500) | Low (₹200 - ₹1,500) | Low (₹500 - ₹4,000) | High (₹15,000 - ₹50,000) | Low (₹50 - ₹500) | Low (₹50 - ₹500) | High (₹20,000 - ₹1,00,000) | Moderate to high (500 - ₹2,000) |
| Propagation Delay      | Low               | Moderate                 | Low                 | High                | Very low (1-10 ms)       | Very low         | Moderate         | High (500 ms)              | 30-100 ms (depends on tech)     |

|                          |           |                     |             |             |                     |                |              |            |                                 |
|--------------------------|-----------|---------------------|-------------|-------------|---------------------|----------------|--------------|------------|---------------------------------|
| <b>Power Consumption</b> | Low (BLE) | Moderate            | Very low    | Very low    | Moderate to high    | Very low       | Low          | High       | Varies (energy-efficient in 4G) |
| <b>Throughput</b>        | 1-3 Mbps  | 54 Mbps to 9.6 Gbps | 20-250 kbps | 0.3-27 kbps | 100 Mbps to 10 Gbps | Up to 424 kbps | Up to 4 Mbps | 1-100 Mbps | 56 kbps (GSM) to 1 Gbps (4G)    |